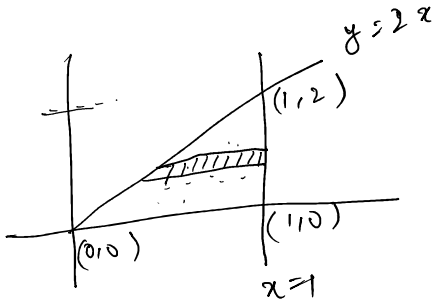
LECTURE 26

28 November 2023 11:33

1) $\int_0^1 \int_0^{2x} dy\, dx = \int_{y=0}^{y=2} \int_{x=1}^{x=y/2} dx\, dy$

$x=0 \qquad y=0$

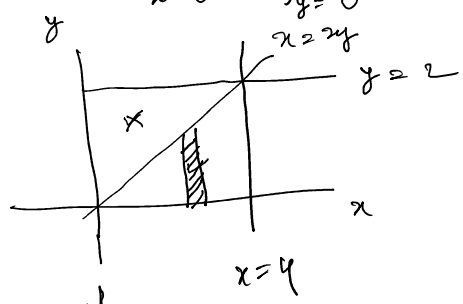
$x=1 \qquad y=2x$



2) $\int_0^2 \int_{2y}^4 dx\, dy = \int_{x=0}^{x=4} \int_{y=0}^{y=x/2} dy\, dx$

$x=4 \qquad x=2y$

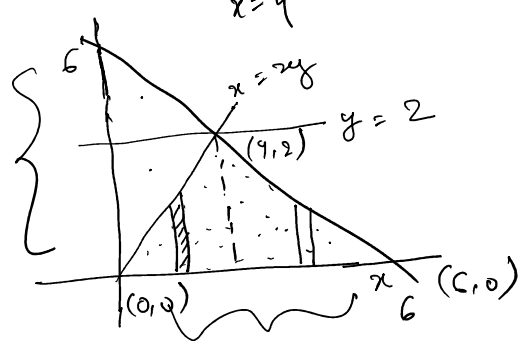
$y=0 \qquad y=2$



3) $\int_0^2 \int_{2y}^{6-y} dx\, dy$

$y=0 \qquad x=6-y$

$y=2 \qquad x=2y$



$x = 6-y$

$x+y=6$

$\frac{x}{6} + \frac{y}{6} = 1$

$\frac{x}{a} + \frac{y}{b} = 1$

$\int_{x=0}^{x=4} \int_{y=0}^{y=x/2} dy\, dx + \int_{x=4}^{x=6} \int_{y=0}^{y=6-x} dy\, dx$

4) $\int_0^4 \int_{\sqrt{x}}^2 \frac{x}{y^5+1}\, dy\, dx = \int_0^2 \int_0^{y^2} \frac{x}{y^5+1}\, dx\, dy$

$x=0 \qquad y=2$

$x=4 \qquad y=\sqrt{x}$

$x=y^2$

$= \int_0^2 \frac{1}{y^5+1} \left(\int_0^{y^2} x\, dx \right) dy$

$= \int_0^2 \frac{1}{\ } \cdot \int \frac{x^2}{\ } \Big|^{y^2} dy$

$x=4 \qquad y=5x$

$x=y^2$

$$= \int_0^2 \frac{1}{y^5+1} \cdot \left[\frac{x^2}{2}\right]_0^{y^2} dy$$

$$= \int_0^2 \frac{y^4}{2(y^5+1)} dy \qquad y^5+1=t$$

$$5y^4\,dy = dt$$

$$= \int_1^{33} \frac{dt}{10 \cdot t} = \frac{1}{10}\left[\log t\right]_1^{33}$$

$$= \frac{1}{10} \log 33$$

Double integration & Volume.

$$V = \iint_R f(x,y)\,dx\,dy = \iint_R f(x,y)\,dA(x,y) = \iint_{R'} f(x,y)\,dy\,dx$$

R: $a \le x \le b$, $g_1(x) \le y \le g_2(x)$

$$\iint_R f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dx\,dy$$

R: $c \le y \le d \qquad h_1(y) \le x \le h_2(y)$

$$\iint_R f(x,y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} dy\,dx$$

1) $\iint_R (e^y + xy)\,dA \qquad R$: Rectangle corner $(3,1)$ & $(4,2)$

$$= \int_1^2 \int_3^4 (e^y + xy)\,dx\,dy = \int_1^2 \left[e^y \cdot x + y\frac{x^2}{2}\right]_3^4 dy$$

$$= \int_1^2 (e^y \cdot 4 + y \cdot 8) - (3e^y + y \cdot \tfrac{9}{2})\,dy$$

2) $\iint_R x^2 y\,dA \qquad R$: $y = 5x$ & $y = x^2$

Ans $\int^1 \int_{-}^{x^2} x^2 y\,dx\,dy$

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Ans $\int_0^1 \int_{\sqrt{x}}^{x} x^2 y \, dx \, dy$